Can News Improve Nasdaq-100 Forecasts?

Evaluating News Inputs and Fine-Tuning Chronos-2

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October 2026

Abstract. Can news improve probabilistic forecasts of QQQ, the ETF tracking the Nasdaq-100? We compare Chronos-2 with and without numeric news summaries, before and after low-rank adaptation (LoRA). A retrospective evaluation across 2022-2025 and three training seeds finds no robust evidence of improvement beyond the market/calendar control: Input-swap tests show that the pretrained model uses little news and fast market inputs mislead it. The largest effect comes from changes in VXN, a measure of expected Nasdaq-100 volatility. We then fine-tune the model using low-rank adaptation (LoRA), which trains small updates while keeping the original weights fixed. Fine-tuning does not change which inputs the model uses; instead, it widens forecast intervals in every week.

1 Introduction

QQQ tracks the Nasdaq-100. We ask whether news helps forecast its returns beyond market data. We evaluate predictive distributions because prior financial evaluations have raised doubts about foundation models' ability to forecast the direction of returns [3]. A quantile is a value below which a stated share of predicted outcomes falls. Our model forecasts 21 quantiles for each of the next five days. A covariate is an extra input series supplied alongside the target return.

RQ1: Do news covariates improve forecasts beyond a market and calendar control? **RQ2:** Does fine-tuning improve a pretrained model with the same inputs? **RQ3:** Does fine-tuning change which inputs the model uses?

We compare a frozen checkpoint with low-rank adaptation (LoRA), which updates small matrices while retaining the original weights. Input perturbations test forecast sensitivity; they do not estimate the causal effect of news on prices.

2 Related work

Temporal Fusion Transformers combine variable selection with multi-horizon quantile forecasts [5]. PatchTST uses series patches and also studies self-supervised pretraining and transfer [9]. Chronos-2 can forecast without task-specific training by sharing information between targets and covariates through group attention [1]. Its quantile outputs and pretrained checkpoint let us vary inputs and compare forecasts before and after adaptation. We did not benchmark TFT or PatchTST.

Rahimikia, Ni and Wang find off-the-shelf time-series foundation models weak on financial excess returns [3]. Our daily QQQ study differs in target, inputs and scoring, so their result motivates direct financial baselines without settling this task.

GDELT provides topic-level news volume and tone [6], Economic Policy Uncertainty counts uncertainty reporting [7], and FOMC-language classification captures policy tone [4]. These represent different hypotheses about relevant news. Average article tone is neither QQQ investor sentiment nor an expected-return estimate. Publication dates align each summary to information available at the forecast origin. LoRA's small trainable matrices make comparisons across sources and seeds affordable on one GPU [2].

3 Data and information availability

The feature table contains 5,201 exchange sessions from January 2006 through September 4, 2026. Appendix B distinguishes the full inventory from selected inputs. News comparisons start in January 2017 because GDELT begins then.

Source family	Selected representation	Role
QQQ and related markets	Returns, gaps, range volatility, volume, momentum, VXN, TLT, HYG	Past only
Rates and macroeconomics	Rate changes, curve slope, real yields, CPI growth	Past only
Event calendar	FOMC, CPI, payroll dates, month-end and option expiry	Known future
EPU and EMU	Log trailing seven-day uncertainty averages	Past only
FOMC statements	Classifier tone, change, smoothed state, elapsed time	Past only
SEC disclosures	Accepted filings and past earnings-event counts	Past only
GDELT topics	Normalized news share and average tone per topic	Past only

Table 1. Data sources. Appendix B lists selected fields, transformations and publication rules.

The target is the daily log return of adjusted QQQ closing prices. The control uses 21 covariates: six QQQ indicators, five market measures, four rates/macroeconomic measures and six calendar fields. All-external adds two uncertainty indices, four FOMC measures, three SEC counts and 12 GDELT measures, giving 42 covariates. Only scheduled calendar fields enter the forecast horizon; other inputs remain in the observed history.

3.1 Publication alignment and news representation

We assign each observation to the first exchange close after publication. July CPI is dated July 1 but released around August 12; joining by reference date exposes six weeks of unavailable information. We compute CPI growth within the ALFRED vintage available at the forecast date. Appendix B records rate releases, filing timestamps, news aggregation and conservative market-data lags.

GDELT topics cover AI, semiconductors, the Federal Reserve, inflation, big-tech earnings and recession. Each contributes news share and average tone, rather than raw headlines [6]. We keep 13 missing sessions per selected GDELT column (0.55%) masked rather than treating them as zero. A pinned RoBERTa classifier scores 170 FOMC statements [4]. The rounds use different feature-table hashes.

4 Model, adaptation and forecasts

Chronos-2 reads up to 512 historical sessions and predicts five daily returns from one information cutoff. Each series is normalized from its observed context and transformed with asinh before patching; no full-sample scaler is fitted [1]. A window contains one history and its following five-session forecast block. After five sessions we append the observed returns and forecast again. Scored blocks do not overlap, although histories do; adapter weights stay fixed during evaluation.

$$W = W_0 + (\alpha/r)BA, \quad r = 8, \quad \alpha = 16 \tag{1}$$

LoRA trains A and B while keeping W_0 fixed. The adapters contain 1,206,912 trainable weights, about 1% of the combined model. B starts at zero [10]. Each run uses the same base checkpoint and retains the final adapter without validation-based checkpoint selection. Appendix A specifies the optimizer, learning-rate schedule, adapted projections, sampling and training loss. Chronos-2 counts target and covariate rows toward the batch [8].

$$r_t = \log(P_t^{adj}/P_{t-1}^{adj}), \qquad \hat{q}_{t+h,\tau},\ h = 1, \ldots, 5 \tag{2}$$

The target is the adjusted-price log return; the model predicts its quantile at level tau for each lead time. These are separate daily returns, not a cumulative weekly return.

5 Training, scoring and the swap test

Property	Round A: source screen	Round B: chronological ladder
Training starts	January 3, 2017	January 3, 2017
Training ends	June 27, 2025	Last session before each test year
Training sessions	2,133	1,259 to 2,012, expanding by year
Scored interval	June 30, 2025-June 26, 2026	Calendar years 2022, 2023, 2024, 2025
Scored days	250 in 50 windows	250 per year, 1,000 distinct days
Completed LoRA fits	18	108 including matched controls
Updates and windows	200 updates, about 1 window each	500 x 8; all-external and matched control 1,000 x 4
Seeds	42; 43/44 for three source setups	42, 43, 44 for every trained setup

Table 2. Training recipes. The rounds change several factors, so their comparison does not isolate a cause.

Round A screens additions to control. Round B is a **retrospective robustness evaluation**: no fold remained independent of model choices. Its eight cumulative sets contain 0, 6, 12, 21, 23, 27, 31 and 42 covariates. They add QQQ indicators, calendar, market/rates, uncertainty, FOMC, GDELT Fed/recession and remaining external fields in that order. Each addition is conditional on earlier groups.

Each fold trains before its test year. The 251, 250, 252 and 250 test sessions yield 50 complete five-session blocks per year, omitting the final one session of 2022 and two of 2024. All-external needs four-window updates to fit memory, so its control uses the same recipe.

5.1 Forecast scores

$$\rho_\tau(u) = \max(\tau u, (\tau - 1)u) \tag{3}$$

$$\mathrm{WQL} = \frac{2 \sum_{t=1}^{N} \sum_{\tau \in Q} \rho_\tau(r_t - \hat{q}_{t,\tau})}{|Q| \sum_{t=1}^{N} |r_t|} \tag{4}$$

N counts scored outcomes; Q contains the 21 levels 0.01, 0.05, 0.10, 0.15, ..., 0.90, 0.95 and 0.99. Pinball loss weights overprediction and underprediction according to the quantile level. WQL averages these losses and normalizes by absolute observed returns. Lower is better; 0.70 does not mean 70% accuracy. Coverage is the fraction of outcomes inside a forecast interval; the 10%-90% interval has nominal coverage 80%. Direction compares the median's sign with the observed return's sign. The zero-mean Gaussian baseline uses the trailing 60-session sample standard deviation at the forecast origin; its zero median counts as up.

5.2 Aggregation and uncertainty

With f indexing years, s seeds and w windows, let L denote mean pinball loss over a window's five outcomes and 21 quantiles. Let a_f denote mean absolute return over all 250 scored outcomes in year f, and a_{fw} its value within window w. We distinguish:

$$C_{fsw} = 2L_{fsw}/a_f, \qquad D_{fsw} = 2L_{fsw}/a_{fw} \tag{5}$$

C is the **fold-normalized contribution** used for model comparisons: its mean over 50 windows equals the year's WQL. D is **window-normalized WQL**, used for swap diagnostics: averaging D gives each window equal weight after its own normalization and need not equal the year's WQL. Their numerical effect sizes are therefore not directly interchangeable.

For paired comparisons, average C over the three seeds within each window, subtract the comparator on the same dates, average the 50 window differences in each year, then give the four years equal weight. The news contrasts and Table 7 use 5,000 circular-block bootstrap draws, independently within each year, with blocks of five consecutive windows (25 sessions). The older pooled fine-tuning intervals use independent-window resampling. Swap intervals average three donor assignments before resampling individual windows. All are exploratory percentile intervals without multiplicity adjustment and condition on the observed seeds. These statistical intervals describe uncertainty in an effect; they are distinct from an 80% return forecast interval.

5.3 Input swaps

A swap replaces one input's complete context with that input's history at another forecast origin, then repeats inference. Known-future calendar values are swapped too; the target and other inputs stay fixed. Positive scrambled-minus-original loss means this perturbation worsened the forecast; negative loss means it improved it. Swaps can break relationships between correlated inputs, and redundant fields can conceal individual effects. This tests sensitivity to a perturbation, not causal influence or whether the model uses a source at all. Donor origins can follow the forecast date, so this remains an offline diagnostic.

6 The first screen and its replication

Round A compares each topic with the same market/calendar control in one seed and one year.

Added news	Fine-tuned WQL	Reduction vs control	Correct direction
None	0.6954	--	140/250
Federal Reserve	0.6865	1.28%	144/250
Recession	0.6868	1.24%	144/250
Semiconductors	0.6875	1.14%	144/250
AI	0.6876	1.13%	143/250
Inflation	0.6881	1.05%	142/250
All external sources	0.6883	1.02%	145/250
Big-tech earnings	0.6887	0.96%	142/250

Table 3. Round A, seed 42. All-external includes non-GDELT sources; all-GDELT alone uses 33 features (WQL 0.6923).

Fed scores 0.6865 against control 0.6954, a 1.28% reduction. The six topic gains span 0.96% to 1.28%; their ranking comes from one seed. Fed direction scores 57.6% and all-external 58.0%, against 57.2% for always up.

Source setup	Mean LoRA WQL	Seed SD	Seeds beating control
Control	0.6936	0.0017	--
EPU / EMU	0.6935	0.0052	1 / 3
All external	0.6894	0.0036	2 / 3

Table 4. Three-seed replication on the same 250 outcomes, not 750 independent days.

Across three seeds, all-external improves mean WQL by 0.60%; its paired interval includes zero. EPU/EMU's seed-42 advantage does not persist at seeds 43 and 44. Round B changes the batch, evaluation period, feature snapshot and library versions; its results cannot isolate the cause of the changed ranking.

7 Does news improve the forecast?

Across four years and three seeds, we find **no robust evidence that the tested news representations improve forecasts beyond the market/calendar control**. The cumulative news additions have small positive mean loss differences, but every paired interval includes zero (Figure 2). This permits modest benefits as well as deterioration; it does not establish that news has no predictive value. The 42-input set worsens mean pretrained WQL by about 0.7% relative to pretrained control.

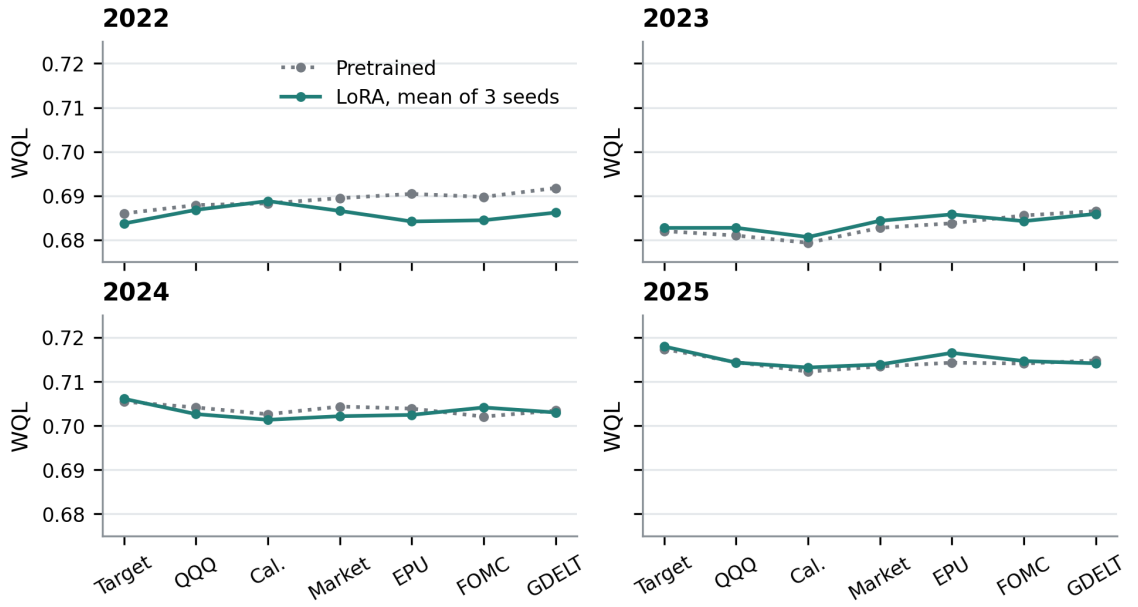


Figure 1. Round B standard ladder: dotted curves show pretrained forecasts; solid curves average the losses of three LoRA seeds. Labels denote cumulative additions. The 42-input matched-recipe comparison appears in Table 5 and Figure 2.

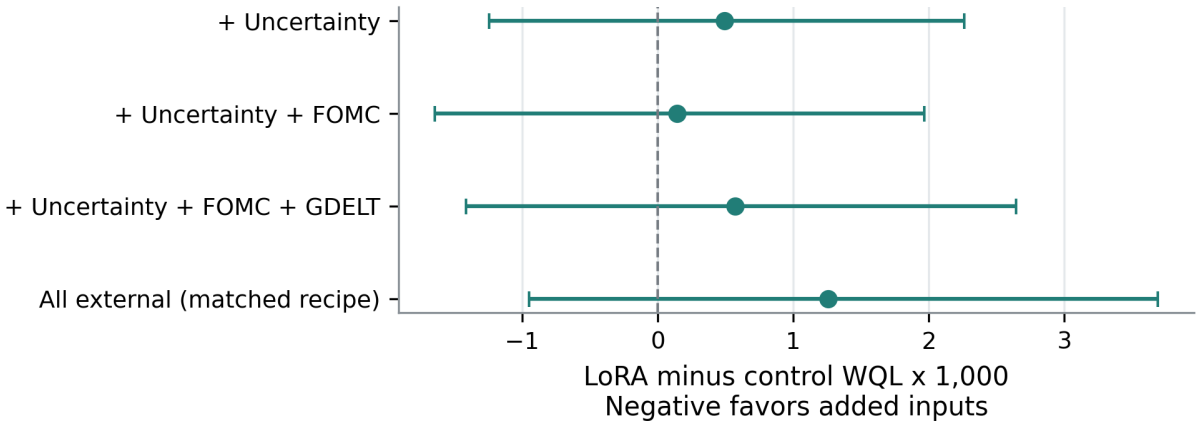


Figure 2. Incremental news effects: seed-mean LoRA WQL minus its recipe-matched market/calendar control, with equal weights for 2022-2025 and exploratory 95% five-window-block intervals. GDELT here adds Fed/recession fields after uncertainty and FOMC. Negative favors the added inputs; these are cumulative comparisons, not isolated source effects.

Model and recipe	2022	2023	2024	2025	Mean
Gaussian, trailing 60 sessions	0.6801	0.6853	0.7133	0.7517	0.7076
Pretrained, control	0.6894	0.6827	0.7043	0.7134	0.6975
Pretrained, all external	0.6977	0.6947	0.7026	0.7199	0.7037
LoRA, all external, 4 x 1,000	0.6904	0.6881	0.7026	0.7147	0.6990
LoRA, control, 4 x 1,000	0.6885	0.6855	0.7034	0.7135	0.6977
LoRA, control, 8 x 500	0.6865	0.6843	0.7021	0.7139	0.6967

Table 5. Annual WQL on the same 250 outcomes per year. LoRA rows average three seeds; Mean gives each year equal weight. The two training recipes sample 4,000 windows but use different update sizes. The Gaussian baseline and pretrained control anchor the absolute scores.

All-external scores 0.6990 against its matched control's 0.6977. The paired difference is +0.00126, with 95% interval [-0.00095, +0.00369]. Against its own pretrained version, its difference is -0.00477, with interval [-0.01013, +0.00032]. Fine-tuning partly offsets the point-estimate loss from extra inputs, without a demonstrated gain beyond control. Across the seven standard input sets, pooled fine-tuned-minus-pretrained WQL ranges from -0.00181 to +0.00037; all corresponding exploratory intervals include zero.

8 Sensitivity of the pretrained model

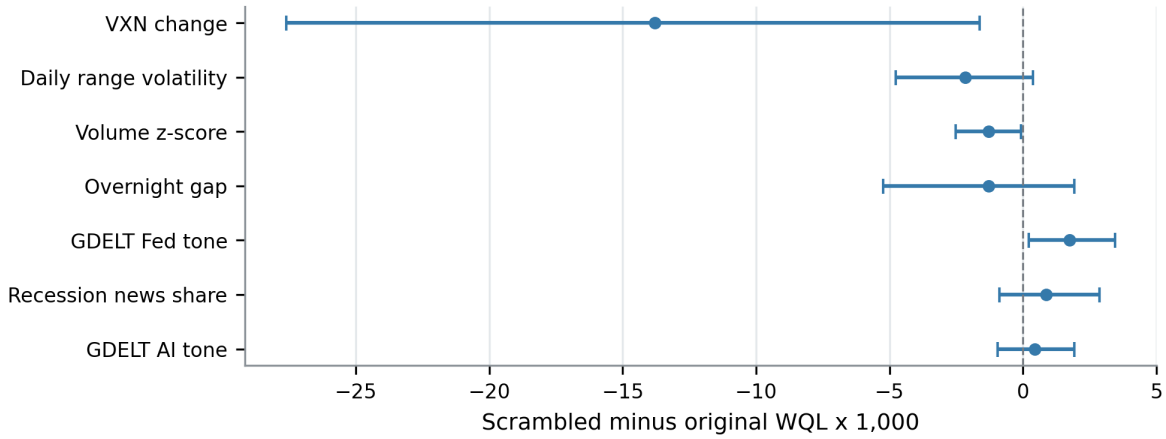


Figure 3. Selected all-external swap effects for June 30, 2025-June 26, 2026, with exploratory 95% individual-window bootstrap intervals. The four most negative mean effects are shown alongside three illustrative news fields: Fed tone, recession share and AI tone. This is a descriptive selection from 42 tests, not a complete ranking. Effects use window-normalized WQL (Section 5.2).

Most individual news-swap effects are small. GDELT Fed tone has a positive mean effect of +0.00175 WQL, but this is one of 42 unadjusted tests and needs replication. Small effects alone cannot distinguish weak use from redundancy or limitations of this perturbation.

VXN measures Nasdaq-100 implied volatility. Scrambling its log change lowers mean window-normalized WQL by 0.0138 with all-external and 0.0134 with the control. For all-external, the median effect is -0.00234; five volatility-shock windows contribute about 94% of the signed total. The strong concentration means that the mean does not describe a typical week. A same-session VXN variant gives a similar result, so the one-session lag alone does not explain the effect. This is evidence of sensitivity to VXN alignment under this diagnostic, not proof that market information is generally harmful. Grouped swaps and feature-removal comparisons would test robustness to correlated inputs and the choice of perturbation.

9 What fine-tuning changes

9.1 Similar importance profiles, unequal effects

We compare six adapted checkpoints: control, control + uncertainty + FOMC, and all-external in 2022 and 2025, all at seed 42. Importance profiles are strongly correlated (0.908-0.978), but correlation is insensitive to a common rescaling and can be dominated by VXN. Removing VXN reduces the range to 0.720-0.953, including 0.720 for all-external in 2025. Thus the profiles are similar in these checkpoints; unchanged feature effects or a general absence of adaptation do not follow. Per-feature changes are included in the reproducibility supplement.

Input setup	2022 all	2022 no VXN	2025 all	2025 no VXN
Market/calendar control	0.948	0.916	0.970	0.953
Control + uncertainty + FOMC	0.963	0.938	0.978	0.953
All external	0.938	0.864	0.908	0.720

Table 6. Pearson correlations between pretrained and adapted mean swap-effect vectors. "No VXN" recomputes the correlation after excluding vxn_log_chg, without retraining either model. Six checkpoint pairs at seed 42 provide a limited diagnostic sample.

9.2 Average widening and calibration

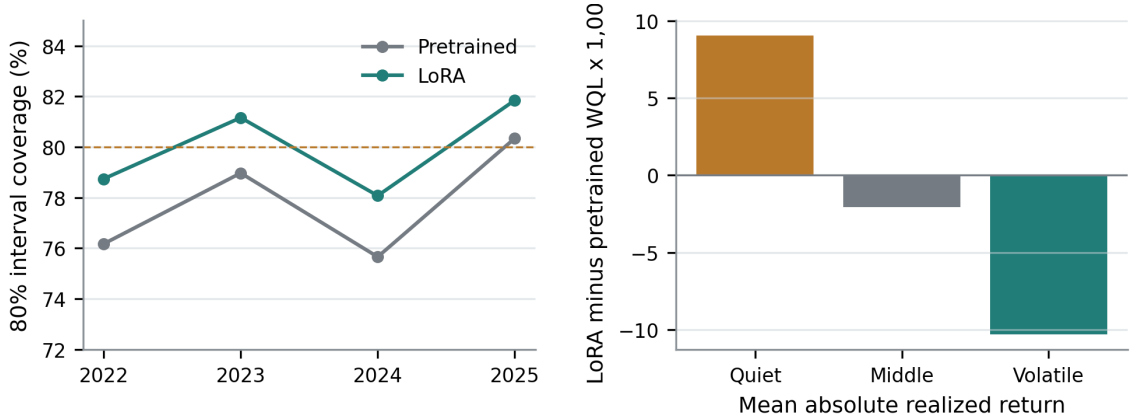


Figure 4. Left: 80% coverage for the seven standard input sets, with equal averages over seeds and setups within each year (84 trained cells). Right: fold-normalized LoRA-minus-pretrained WQL, averaging seeds and input sets 3-6 within each of 200 distinct five-session blocks before dividing them into 67 quiet, 66 middle and 67 volatile blocks. Movement is mean absolute realized return within the block; grouping is retrospective.

Across all 108 trained cells, including the matched-recipe runs, mean 80% coverage rises from 77.69% to 80.09%. The equally weighted mean of each cell's fine-tuned/pretrained mean-width ratio is 1.0780. This is a 7.80% average increase, not a ratio of pooled widths. Mean width increases in 106 of 108 cells; changes range from -0.83% to +14.55% (Figure 5). A cell is one input setup, year, seed and training recipe, not an independent market history.

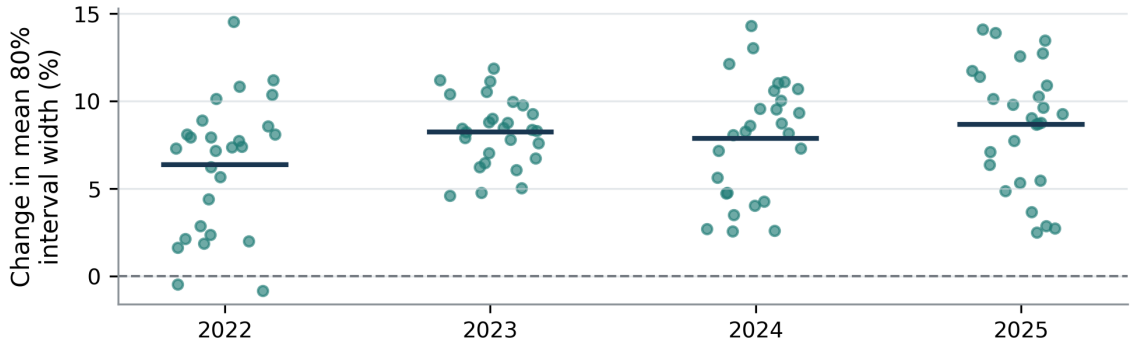


Figure 5. Change in each trained cell's mean 10%-90% interval width: 27 cells per year, 108 total. Each point averages that cell's 250 daily forecasts; bars show the yearly mean of these ratios. This does not show individual-week changes or establish whether widening responds to forecast-time conditions.

In the separate 200-block analysis, WQL rises by 0.0090 in the quiet third and falls by 0.0103 in the volatile third. These outcomes suggest that broad widening can help in volatile periods and cost in quiet ones. They do not establish a forecast-time rule for when to widen. The local Round B export lacks daily trained quantiles, so weekly width changes cannot be reconstructed from it.

Contrast	Mean WQL difference	Five-window-block 95% interval
All-external LoRA minus matched control	+0.00126	[-0.00095, +0.00369]
All-external LoRA minus own pretrained	-0.00477	[-0.01013, +0.00032]
Control LoRA minus Gaussian	-0.01088	[-0.02269, -0.00106]

Table 7. Paired contrasts with seeds averaged, equal year weights and 5,000 circular-block bootstrap draws of five windows. Negative differences favor the first model; intervals remain exploratory. The Gaussian comparison uses the standard eight-window control recipe.

9.3 A simple interval-rescaling comparison

To test whether adaptation adds value beyond changing spread, we rescale pretrained control quantiles around their median:

$$\hat{q}'_{t,\tau} = \hat{q}_{t,0.5} + a(\hat{q}_{t,\tau} - \hat{q}_{t,0.5}), \quad a \geq 0 \tag{6}$$

For each test year in 2023-2025, fit a nonnegative scale on all preceding years available from 2022, minimizing equal-year WQL. Freeze the scale during testing and preserve the median. This retrospective comparison covers 750 outcomes; 2022 supplies calibration only. Appendix A details the protocol.

Test year	Calibration years	Scale	Pretrained WQL	Rescaled WQL	LoRA WQL
2023	2022	1.075	0.6827	0.6845	0.6843
2024	2022-2023	1.014	0.7043	0.7040	0.7021
2025	2022-2024	1.040	0.7134	0.7131	0.7139

Table 8. Pretrained control rescaling uses only earlier calibration years. LoRA averages three seeds of the standard eight-window recipe.

Mean LoRA-minus-rescaled WQL is -0.00047, with exploratory five-window-block 95% interval [-0.00377, +0.00253]. This establishes neither a reliable LoRA advantage nor equivalence. Rescaling itself does not improve pretrained WQL every year.

10 Discussion

The experiments find no robust news benefit (RQ1), uncertain WQL gains from LoRA despite better coverage (RQ2), and similar importance profiles in six checkpoints (RQ3). The standard LoRA control beats the Gaussian by 0.01088 mean WQL but loses in 2022. Round A direction scores of 57.6%-58.0% barely exceed always-up's 57.2%; trading value is unestablished. No tree or learned volatility model was tested.

Round B changes batching, dates, snapshots and library versions together, preventing attribution to one change. Repeated seeds quantify optimization variability on shared outcomes, not independent temporal replication.

10.1 Limitations

All folds share one ETF and were exposed to model choices. Chronological training does not remove retrospective specification bias. Exact QQQ pretraining exposure remains unverified; calendar overlap alone does not establish contamination [1]. Revised data, retrospective Fed classification and hindsight-selected SEC firms limit historical realism.

Intervals are exploratory, unadjusted for multiplicity and conditional on three seeds. Correlated inputs and unrealistic swaps limit importance interpretation. Numeric news summaries may omit event timing and relevance. Aggregate coverage does not establish conditional calibration; missing daily trained quantiles prevent weekly width checks. Rescaling covers only control and three test years.

10.2 Future work

Prioritize a frozen prospective evaluation, four-year rescaling, and learned volatility/quantile baselines. Export daily quantiles and repeat grouped swaps and feature-removal tests with matched recipes. Explore news embeddings, unexpected coverage changes, headline novelty, and Nasdaq-100 company relevance, followed by daily forecast origins and full-weight adaptation.

11 Conclusion

The tested news summaries show no robust gain beyond control. LoRA widens intervals on average and improves aggregate coverage, with uncertain WQL benefit. VXN sensitivity and similar importance profiles warrant further tests. These findings concern one ETF, representation, horizon and training budget.

Code availability. Source code is available at https://github.com/EladKulman/QQQ-Time-Series-Predictor. Reproducibility details are in Appendix A.

References

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Appendix A Execution record and reproducibility

A.1 Training recipe

Saved runs use learning rate 1e-5, context cap 512, five-session targets and the final checkpoint. Chronos 2.3.2 supplies fused AdamW, linear learning-rate decay, zero warm-up and one-step gradient accumulation [8]. These settings are recovered from the versioned training implementation, not serialized optimizer states. Training minimizes masked quantile loss on normalized targets; evaluation WQL uses log-return units.

LoRA uses rank 8, alpha 16, dropout 0 and no trained bias. Adapted modules are self-attention q/k/v/o and the output patch embedding's output layer. Round A adapter files confirm these settings; Round B uses the same library defaults. Origins are sampled uniformly with replacement within training data, with at least five history sessions and at most 512. Thus 4,000 window draws need not be distinct. Table 2 gives update counts; with k covariates, the series-row batch equals the requested window count times (k + 1).

A.2 Execution and artifacts

Round	Slurm jobs	Completed work
Infrastructure	869927	Five LoRA updates, 15 outcomes, adapter reload
A: source screen	869989; 871482; 874192	18 fits, 200 updates per fit
B: ladder	896510; 896540; 896541	84 fits, seven rungs x four years x three seeds
B: matched recipe	896751; 896752	24 fits, all-external and control, 4 x 1,000
B: diagnostics	896501; 896762	32 pretrained cells; six paired checkpoint perturbation tests

Table 9. The 126 substantive LoRA fits comprise 18 first-round fits plus 108 later fits. Additional pretrained exploration and perturbation inference ran on MPS; these are not additional training seeds.

Both rounds pin model revision `29ec3766d36d6f73f0696f85560a422f50e8498c`. Round A data SHA-256 is `e0c8d48b38059eb70d9f91931bff68678dbcc46aeadab1502ded1b9ee11df037`; Round B is `b3beb046933b15f1f75c098bfa6df30af4106395302ea4d8b68711322a262e34`. Training used an NVIDIA RTX 2080 Ti (11 GB) through Slurm. Both GPU runtimes record Chronos 2.3.2 and PyTorch 2.11.0+cu128. Transformers/PEFT change from 5.16.1/0.20.0 to 5.17.0/0.21.0. Later metadata records dirty working trees, making source hashes material to reproduction.

Round A predictions and adapters reside in `models/news-ablation-869989/`. Round B local exports in `docs/results/walk_forward/` include per-window loss summaries and 140 cell metadata/metric pairs (108 trained, 32 pretrained-only). Full later adapters and per-day trained predictions are not in that local export. The original analyzer checks cell metrics against window means to 3.4e-16 and records source hashes in `output/pdf/qqq_report_support/evidence.json`. Current configuration files have since changed; use the saved run metadata to recover historical recipes.

The accompanying `report/README.md` gives analysis and build commands. The `report/review_analysis/` supplement contains all Round B run metadata, unrounded contrasts, feature-effect changes, widths and recalibrated predictions; `summary.json` records input hashes and bootstrap settings. Run `scripts/refresh_report_review.py`, then `scripts/build_report.sh`. These paths refer to the accompanying repository. Hashes alone cannot recover unarchived dirty source trees, and missing trained artifacts still limit full reproduction.

A.3 Recalibration protocol

The pretrained control export in `docs/results/pretrained_ladder/` matches Round B's checkpoint, data, feature-role hash, horizon and quantile grid; CUDA window scores agree within 5.6e-7 WQL. The analysis exactly minimizes the convex piecewise-linear pinball objective over nonnegative scales using preceding years only. It fixes the scale throughout each test year. Intervals resample five-window blocks, weight 2023-2025 equally, and condition on the fitted scales; they omit uncertainty from refitting calibration. No Chronos model was retrained for these additions.

Appendix B Exact inputs and source definitions

Definitions precede model scaling. Logs are natural; rolling windows count exchange sessions. "Past" denotes context-only inputs; "Future" also permits scheduled horizon values. Publication alignment follows Section 3.1.

Column	Meaning / transformation	Role
overnight_gap	QQQ log(open / previous close).	Past
parkinson_vol_1d	Daily range volatility: abs(log(high / low)) / sqrt(4 log(2)); not annualized.	Past
parkinson_vol_22d	22-session mean of daily range volatility (minimum 15 observations).	Past
volume_z_20d	20-session z-score of log QQQ volume (sample standard deviation; minimum 15 observations).	Past
mom_21d	QQQ log(close / close 21 sessions earlier).	Past
dist_200dma	QQQ log(close / 200-session mean close); minimum 150 observations.	Past
vxn_log_chg	Daily log change in lagged Nasdaq-100 implied volatility (VXN).	Past
vxn_minus_vix	Lagged VXN minus S&P 500 implied volatility (VIX), in index points.	Past
vix_term_ratio	Lagged VIX / VIX3M: one-month / three-month implied volatility.	Past
tlt_log_ret	Daily log return on the long-term Treasury bond ETF TLT.	Past
hyg_log_ret	Daily log return on the high-yield corporate bond ETF HYG.	Past
d_dgs2_bp	Change in available two-year Treasury yield, in basis points.	Past
t10y2y_lag1	Available ten-year minus two-year Treasury yield, in percentage points.	Past
d_dfii10_bp	Change in available ten-year real Treasury yield, in basis points.	Past
cpi_yoy	Released-vintage CPI / CPI 12 months earlier - 1; fraction.	Past
is_fomc_day	1 on a scheduled FOMC decision day; 0 otherwise.	Future
is_cpi_day	1 on a CPI release day; 0 otherwise.	Future
is_nfp_day	1 on a nonfarm-payroll employment release day; 0 otherwise.	Future
days_to_fomc	Calendar days to the next scheduled FOMC decision; 0 that day.	Future
days_to_month_end	Session window: -3 to -1 before month-end, +1 to +3 after; 0 otherwise.	Future
is_opex_week	1 in monthly option-expiration week (holiday-adjusted third Friday); 0 otherwise.	Future

Table 10. The 21 control covariates: six QQQ, five market, four rates/macroeconomic and six calendar fields. Fifteen are past-only and six are known-future.

Each topic adds two past-only fields: `gdelt_TOPIC_news_share` (matching articles / normalization count) and `gdelt_TOPIC_avg_tone` (mean article-tone score). TOPIC is `ai`, `semiconductor`, `fed`, `inflation`, `big_tech_earnings` or `recession`.

GDELT topic	Query before the English-language filter
ai	("artificial intelligence" OR "generative AI" OR "machine learning" OR OpenAI OR Nvidia)
semiconductor	(semiconductor OR chipmaker OR Nvidia OR AMD OR TSMC OR ASML OR Intel)
fed	("Federal Reserve" OR FOMC OR "interest rates" OR "rate hike" OR "rate cut")
inflation	(inflation OR CPI OR "consumer prices" OR "core inflation")
big tech earnings	(Apple OR Microsoft OR Amazon OR Alphabet OR Meta OR Nvidia) (earnings OR revenue OR guidance)
recession	(recession OR "hard landing" OR "economic contraction")

Table 11. GDELT queries use sourcelang:eng and no timeline smoothing. Raw article counts are not selected inputs.

Column	Meaning / transformation (past-only)
epu_log	Log trailing seven-calendar-day Economic Policy Uncertainty mean (minimum four observations), aligned to sessions.
emu_log	Same transformation for the Equity Market Uncertainty index.
fomc_net_hawkish_last	Latest statement: hawkish minus dovish sentence share (-1 to +1), carried forward.
fomc_net_hawkish_ewma	Smoothed net-hawkish score; half-life three statements, carried forward.
fomc_tone_change_last	Latest between-statement change in net-hawkish score, carried forward.
days_since_fomc	Calendar days since the latest source FOMC statement.
ndx_earnings_count	Observed earnings-related 8-K count (Item 2.02), aligned to the first eligible close.
sec_8k_count	Accepted 8-K count at the eligible exchange close.
sec_total_filings	Accepted 10-K, 10-Q, 8-K, 20-F and 6-K count at the eligible close.

Table 12. Nine past-only external fields; with 12 GDELT fields and 21 controls, all-external has 42 covariates.

SEC counts cover AAPL, MSFT, NVDA, AMZN, GOOGL, META, AMD, INTC, TSM and AVGO, including configured predecessors. They record accepted filings, not an advance or complete Nasdaq-100 earnings calendar.

EPU and EMU derive from the policy uncertainty data family [7]. The FOMC model used in the pipeline is LorenzoAleCon29/roberta-large-fomc-hawkish-dovish, revision f4759d4ad3f1182f81d87e47ba603261740d36cf. Missing observations use the model's observation mask. The array-based Chronos interface preserves exchange-session order without fabricating holiday rows [8].

B.1 Availability and inventory details

The 77-column inventory includes the date, target, 55 past-covariate fields, 14 eligible future-calendar fields and six raw fields. Prices from 2005 warm up indicators. H.15 rates use the 16:15 release on the next publication business day, then the next eligible exchange close. SEC filings use UTC acceptance timestamps, including early closes. Complete daily GDELT aggregates shift by one calendar day, roll to exchange sessions and average weekend observations. CBOE-derived inputs receive a one-session lag. Only scheduled calendar information enters the future channel; future news, prices and realized returns remain masked.